SoundReg: working brief (problem, math, experiment)

1. Problem

From a short multichannel window $\mathcal{A}$ of one ego-mounted microphone array (audio only, no vision at inference), output the set of road users currently making task-relevant sound, each as (class, range, azimuth) in the bird's-eye-view (BEV) plane:

$$\mathcal{S} = \{(c_i, r_i, \theta_i)\}_{i=1}^{N}, \qquad N \text{ unknown and variable.}$$

This is the object list a planner already consumes from LiDAR and camera, not the single direction that classical localization or SELD produce. Sound is useful because it is not line-of-sight bound (sources audible around corners, before they are visible). LiDAR gives accurate positions for in-sight objects and is used as the training label there.

2. Idea (one sentence)

Masking is the acoustic analog of occlusion: a loud source hides weaker concurrent sources in the time-frequency (TF) bins it dominates, so generate objects loudest first (dominance order) and condition each one on those already emitted. The ordering is an inductive bias that should help a finite-capacity model recover low-SNR masked sources; expect the benefit to be largest on hard cases and in the small-data regime, and to shrink with more data.

3. Signal model and the quantities that matter

Stacked-channel STFT of the mixture, with per-source array images $\mathbf{x}_i$ and noise $\mathbf{n}$:

$$\mathbf{X}(t,f) = \sum_{i=1}^{N} \mathbf{x}_i(t,f) + \mathbf{n}(t,f), \qquad E_i = \sum_{t,f} \|\mathbf{x}_i(t,f)\|^2. \tag{1}$$

Difficulty of source j seen against all others (what a parallel detector faces), and after a set C of sources is accounted for:

$$\mathrm{SNR}_j(\varnothing) = \frac{E_j}{\sum_{i \neq j} E_i^{(j)} + E_n^{(j)}}, \qquad \mathrm{SNR}_j(C) = \frac{E_j}{\sum_{i \notin C \cup \{j\}} E_i^{(j)} + E_n^{(j)}} \;\geq\; \mathrm{SNR}_j(\varnothing), \tag{2}$$

where $E_i^{(j)}, E_n^{(j)}$ are interferer/noise energy in j's TF support. The useful quantity is the *headroom* $\mathrm{SNR}_j(C) - \mathrm{SNR}_j(\varnothing)$: zero when nothing masks j, large when C holds j's maskers. $\mathrm{SNR}_j(C)$ is an oracle (perfect removal of C); the model only approaches it by conditioning on emitted tokens. Generation order is set by a dominance score (steered-response power), emitted in descending d_i:

$$d_i = \sum_{t,f} \left| \mathbf{w}(r_i, \theta_i)^{\mathsf{H}} \mathbf{X}(t,f) \right|^2. \tag{3}$$

4. Model

Each object is three tokens: class c_i, quantized range t_i^r, quantized azimuth t_i^θ (separate vocabulary per axis; azimuth finer than range). A scene in dominance order π and its likelihood:

$$O_\pi = [\,\texttt{[BOS]}, (c, t^r, t^\theta)_{\pi(1)}, \ldots, (c, t^r, t^\theta)_{\pi(N)}, \texttt{[EOS]}\,], \tag{4}$$

$$p_\Theta(O_\pi \mid \mathcal{A}) = p_\Theta(\texttt{[EOS]} \mid h_{N+1}) \prod_{i=1}^{N} p_\Theta(c_i \mid h_i)\, p_\Theta(t_i^r \mid c_i, h_i)\, p_\Theta(t_i^\theta \mid c_i, t_i^r, h_i), \tag{5}$$

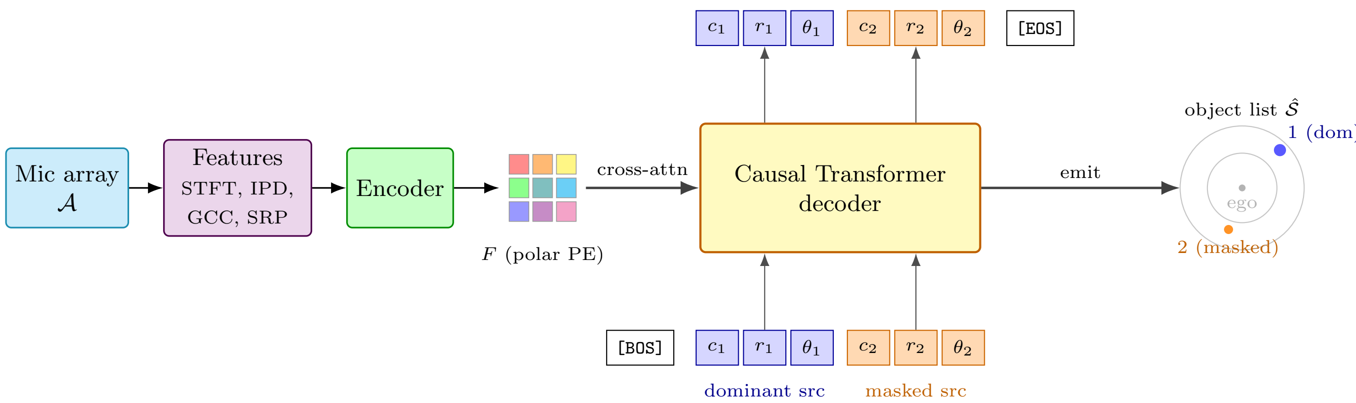

Figure 1: SoundReg. Audio is encoded once into scene features F; a causal decoder cross-attends to F and emits sounding objects one at a time in dominance order. The dominant source (blue) is emitted first; conditioning on it is intended to let the decoder recover the masked source (orange) that a parallel detector, working from the full mixture, tends to drop.

where h_i summarizes the scene features F and the objects already emitted. Train by teacher forcing with one cross-entropy:

$$\mathcal{L} = -\sum_{i=1}^{N} \big[\log p_\Theta(c_i \mid h_i) + \log p_\Theta(t_i^r \mid c_i, h_i) + \log p_\Theta(t_i^\theta \mid c_i, t_i^r, h_i) \big] - \log p_\Theta(\texttt{[EOS]} \mid h_{N+1}). \tag{6}$$

Pieces to implement:

- Encoder: features $\mathcal{A} \to$ STFT, inter-channel phase difference (IPD), GCC-PHAT, optional SRP map on the polar grid $\to F$ with a polar positional encoding (separate range and azimuth embeddings).
- Decoder: causal Transformer, cross-attends to F, token-type masking (class steps emit class tokens, location steps emit location tokens).
- Cardinality: `[EOS]` ends the sequence (no fixed query count, no threshold).
- Inference: greedy/beam for main numbers; sample S sequences and cluster for existence/uncertainty.

5. Data and ground truth you need

- Real array+LiDAR driving data (all line-of-sight). Decide and record: array geometry/channels, polar grid $(N_\theta, N_r, r_{\max})$, train/val/test split.
- Ground truth = objects *actually emitting* task-relevant sound, with class and position. Sound-activity must be annotated independently of the detector (else the evaluation is circular). Position from the LiDAR track.
- Per-source SNR (for stratifying recall): exact in simulation; on real LOS data estimate by beamforming toward each labeled source.
- NLOS: (i) simulated NLOS scenes with known positions mixed into training of every set-prediction model, (ii) a small hand-annotated real NLOS test set. Scoped demo, not a benchmark.

6. Experiment (run in this order)

Baselines (parallel): GCC/SRP peak picking; SELD-style class+DOA; polar heatmap + peak extraction; query-set predictor (DETR / Sound3DVDet style). SoundReg is the only sequential one.

Metrics: match prediction to GT by class + polar-distance threshold; report F1, cardinality error ($|\hat{N} - N|$), range/azimuth MAE. Report mean ± std over several seeds.

Headline test (do this first). Recall stratified by per-source SNR and by number of concurrent sources, SoundReg vs each baseline.

- Prediction: SoundReg's recall advantage *grows* as SNR drops and overlap rises.

- Falsifier: a flat advantage. If the gap does not grow at low SNR, the masking idea is wrong, stop here.
- Watch the opposite failure: the decoder can fire `[EOS]` early and *drop* quiet masked sources, which shows up as *worse* low-SNR recall. Monitor `[EOS]` calibration; also report a length-controlled decoding variant.

Ablations (each isolates one thing).

- Order: dominance vs distance vs azimuth-sweep vs random vs oracle masking-order (upper bound).
- Conditioning: full vs context-masked decoder (predict each object without seeing previously emitted objects). This is the cleanest test that conditioning, not tokenization, does the work.
- Dominance estimator: beamformer power vs oracle energy vs SNR.
- NLOS: train with vs without simulated NLOS data; compare NLOS detection rate against the query baseline given the *same* NLOS targets.
- Soft target: one-bin Gaussian vs hard label.

Second prediction (data scaling). The advantage should shrink as training data grows (it is an inductive bias, not an information gain). Run a small data-fraction sweep against the query baseline.

7. The data on hand, and what to prepare

Each sample (one `token`) carries:

```
items[token] = {
  "stft":     ndarray (64, F, T)   # 32 mics; complex split to 64 real ch; encoder input
  "audio":    ndarray (T, 32)      # optional raw multichannel
  "ann":      [ {"center":[x,y,z], "wlh":[w,l,h]}, ... ]  # sounding objects only
  "lidar":    ndarray (N, 3)       # box-position source; not used at inference
  "velocity": {lin_x,lin_y,lin_z, ang_x,ang_y,ang_z}     # ego motion (optional)
  "timestamp", "scene_name"
}
```

The annotated boxes are the sounding objects; silent objects are simply not annotated, so there is nothing to detect for them. A single class is used, so the class token c_i in Eqs. (4) to (6) is constant and drops out: a target token is just (r, θ). Microphone positions come from a separate XML file (the array geometry).

Preparation. Each sample becomes one (features, ordered target sequence) pair:

1. Mic geometry: parse the XML into microphone positions $\in \mathbb{R}^{M\times 3}$. One-time.
2. Polar grid: fix $r_{\max}$, N_r, N_θ. This is the target vocabulary.
3. Steering vectors: precompute $w(r,\theta)$ per grid cell and frequency from the geometry. One-time.
4. Features: from `stft`, build log-magnitude and inter-channel phase difference (optionally GCC-PHAT). Encoder input.
5. Targets: each `ann` center $[x, y]$ gives $r = \sqrt{x^2+y^2}$, $\theta = \operatorname{atan2}(y,x)$, quantized to a (r,θ) bin.
6. Order: dominance $d_i = \sum_{t,f} |w(r_i,\theta_i)^{\mathsf{H}}\mathbf{X}(t,f)|^2$ per box; sort high to low; prepend `[BOS]`, append `[EOS]`.
7. SNR (evaluation only): beamform toward each source, estimate its SNR, store it with the sample to stratify recall.

After step 7 each sample is $[$`[BOS]`$, (r,\theta)_1, \ldots, (r,\theta)_N,$ `[EOS]`$]$ paired with its features, and the model trains directly on it.